

Question and design

Can sequence-only inference return both observed folds of a fold-switching protein? We ran the exp277 contacts-v1 checkpoint on 67 non-capped pairs. The primary test set has 29 pairs with at least 98% identity, held apart by sequence group from 15 development pairs. Seventeen of those 29 are literally identical. Every method returned at most 16 ranked maps chosen without reference contacts or structures.

How the 29-pair test set was selected

The source was 93 literature-curated pairs in the NCBI AF2 benchmark Table S1. The reference-defined funnel retained 68 premise-valid pairs, 65 with at least 98% sequence identity, 45 with at least 10 switching-region contacts unique to each fold, and 44 after excluding one context-capped pair. Sequence grouping followed by deterministic SHA-256 ordering assigned 15 pairs to development and 29 to the held-out test set. The cap exclusion used completion feasibility; no fold-recovery quality outcome entered selection. Of the final 29, 17 are exact-sequence pairs and 12 differ by 1–5 substitutions.

The primary comparison is negative

At 100 shared root plus 100 arm rollouts per protein, 10-contact clustered branching improved mean minority-fold enrichment by only 0.013 over independent sampling. The paired 95% interval is -0.014 to +0.043; mean minority recall was essentially unchanged. Blind dual-contact hits were 3/29 versus 2/29. The extra hit failed a 3D check. On the 17 exact-sequence test pairs, both methods had one contact hit and the paired interval still spans zero. We cannot claim a reliable search improvement.

What was actually discovered

For 3j7w/3j7v, both independent and branching top-16 shortlists included maps that reconstructed distinct switching-region structures. With the Fold1 sequence input, region-touch IDDT against (Fold1, Fold2) was (0.477, 0.206) and (0.356, 0.702) for the branching shortlist; the independent shortlist also favored the respective folds. True-contact controls showed comparable separation. This is an existence proof for blind inference discovery, not a branching advantage.

Where the search is limited

Four of seven branch-10 oracle dual-contact hits were present among 200 candidates but missed by the blind 16-map selector. In a separate plain-sampling run, blind dual hits were 2, 3, and 3 at 100, 200, and 500 rollouts; the oracle pool reached 9 by 200. The first priority is better reference-free ranking of coherent minority maps. The all-rollout structural analysis below also shows that the contact-level proxy admits many candidates that do not reconstruct near the nominal fold.

What the 500-rollout oracle means

The oracle inspects all 500 independent maps against both reference contact maps. A dual hit needs one Fold1-like and another Fold2-like map; it does not mean the blind selector found either. Among 29 held-out primary pairs, 19 had any Fold1-like map, 13 any Fold2-like map, and 9 had both. The same counts for the 17 exact-sequence test pairs were 12, 7, and 5. Across all 67 non-capped pairs, 31 had both. The prespecified 25% contact-recall screen is permissive: raising recall to 50% reduces held-out dual coverage to 2/29. The later all-rollout structural run confirms that contact-screen coverage is not a count of recovered three-dimensional folds.

Do another 500 iid draws help?

Yes, modestly for oracle contact-pool coverage. On the same 29 primary test proteins, the count with both fold-like contact modes rises from 9 at 500 draws to 11 at 750 and stays 11 at 1,000. Neither-mode proteins fall from 6 to 5. The two new dual hits first appear at draws 509 and 587, and both are exact-sequence pairs. At the stricter 50% recall cutoff, the dual count stays 2/29. These are reference-aware screening hits in the full pool. Folding every map separately finds no new strict structural dual after draw 249.

Every iid map was folded separately

We ran 29,000 individual iid contact maps plus 87 true-Fold1, true-Fold2, and no-contact controls through Helico: one sample, six cycles, seed 42, no MSA. All 29,087 predictions and PDBs completed. Both true-contact controls pass for 17/29 proteins. Under the exploratory strict screen—global Kabsch GDT-TS at least $\max(0.35, 90\%$ of the corresponding control) plus switching-region advantage at least 0.10—the number with both modes is 0 at 100 draws, 1 at 200, and 2 from 500 through 1,000. Both dual cases are in the 17-protein exact-sequence subset.

The structural positives are not interchangeable

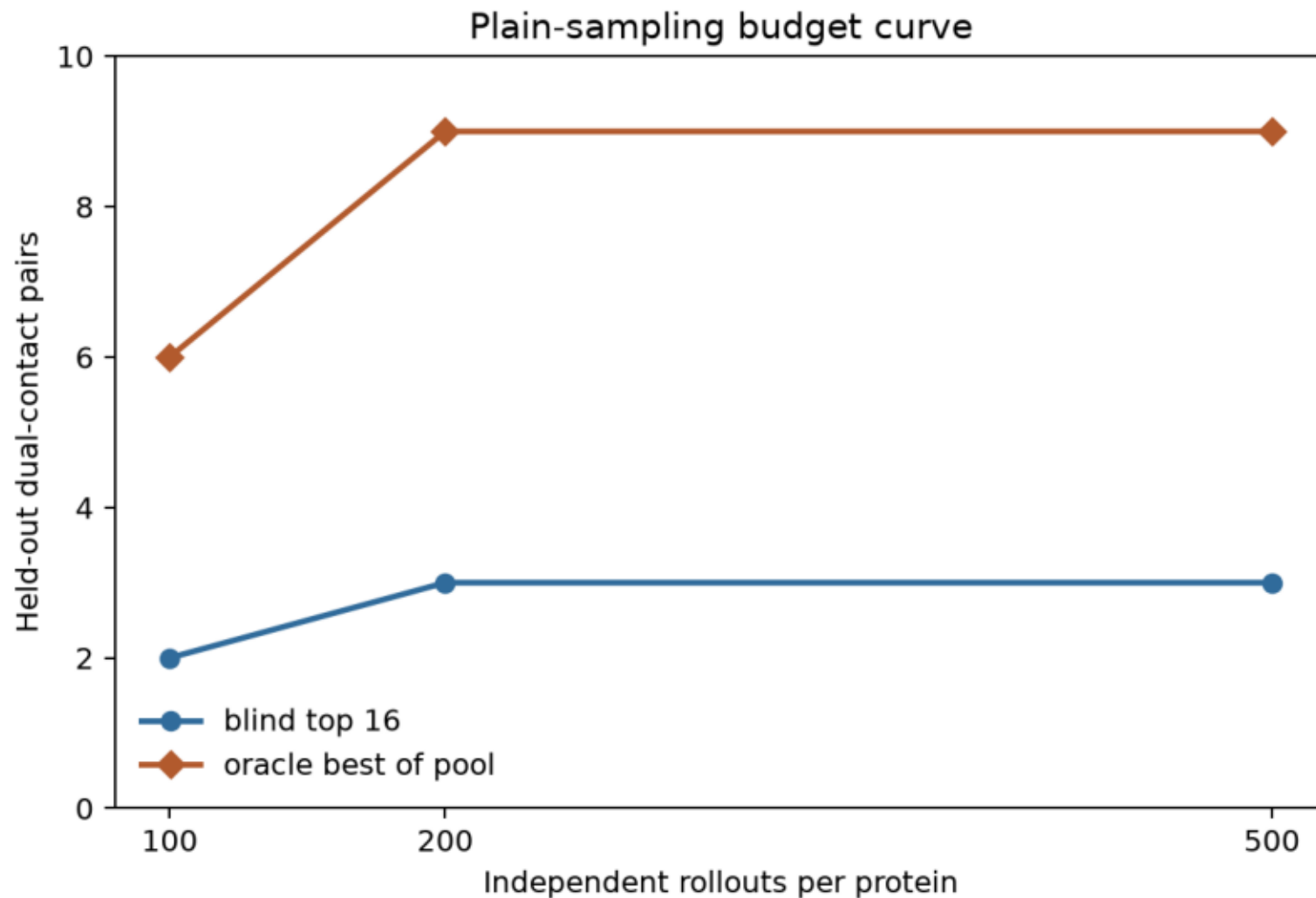
3j7w/3j7v is the cleanest contact-driven example: both controls pass, the no-contact baseline matches neither, and both modes appear by draw 144. 2lel/2k0q passes the strict control-relative screen by draw 249, but its no-contact baseline is already Fold2-like; search contributes the Fold1-like mode. 1qs8/1miq is the strongest near-threshold case: its candidates reach global GDT-TS 0.855 and 0.758, but the latter is 84.7% of its unusually strong control and misses the 90% rule.

Binary structural coverage is threshold-sensitive

Using 80% rather than 90% of the true-contact control gives 3/29 dual proteins by adding 1qs8/1miq; requiring 100% gives 1/29. Fixed absolute GDT-TS cutoffs of 0.35, 0.50, and 0.60 with the same 0.10 region margin give 8, 3, and 1. The raw scores and sensitivity grids are the result to trust. Contact and structural preference correlate (Spearman rho 0.705), but the contact screen reports 11/29 duals while the strict structural screen reports 2/29. Missing modes are inconclusive for the 12 proteins that fail at least one true-contact control.

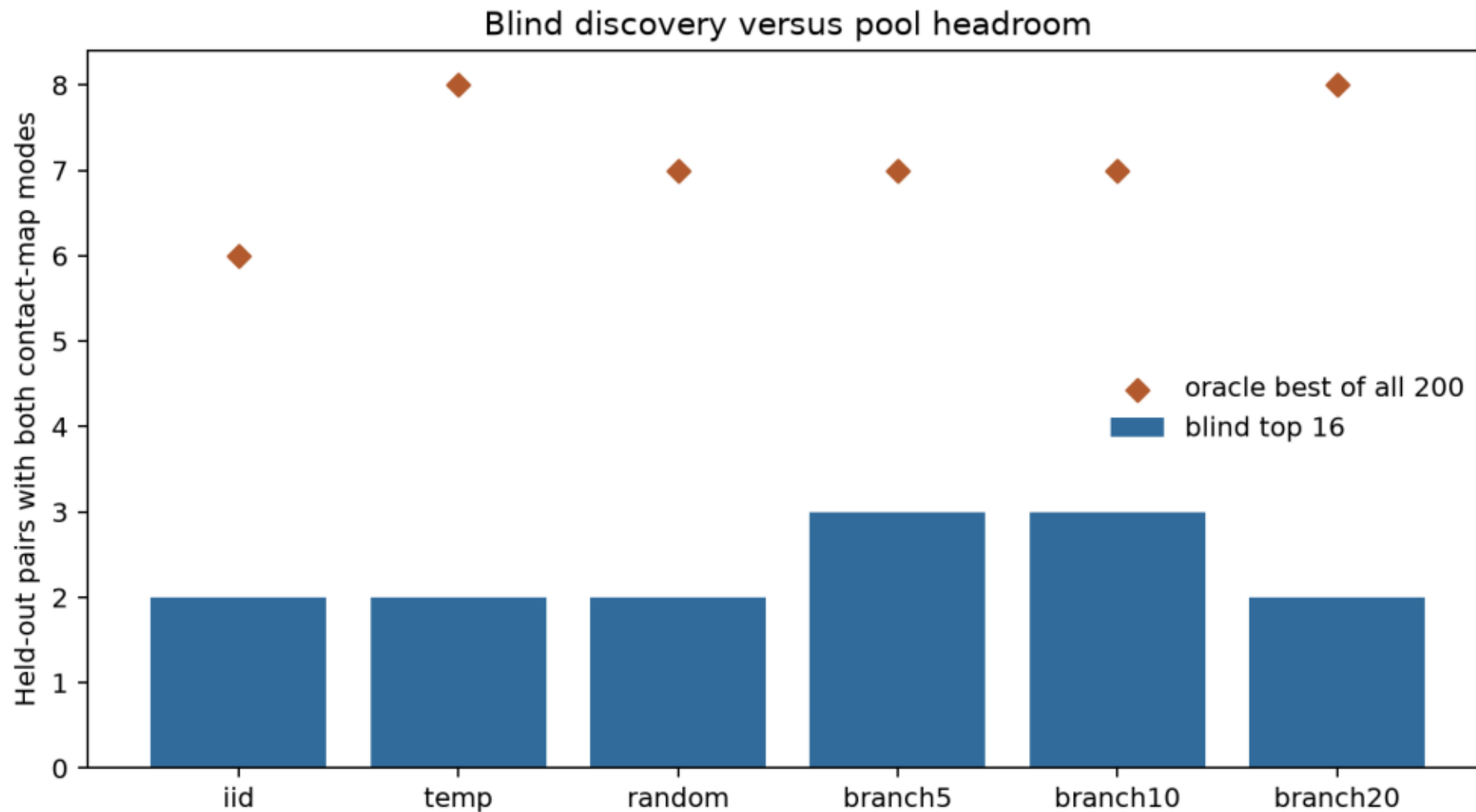
budget_curve.png

A separate 500-rollout run, evaluated at its first 100, 200, and 500 candidates. The blind selector returns 16 maps at each budget; the oracle line is diagnostic and cannot be used at inference time.



dual_contact_hits.png

Bars count pairs meeting the prespecified contact-level dual-hit criterion among 16 blindly selected maps. Diamonds show the oracle ceiling in the same candidate pool, which cannot serve as an inference method.

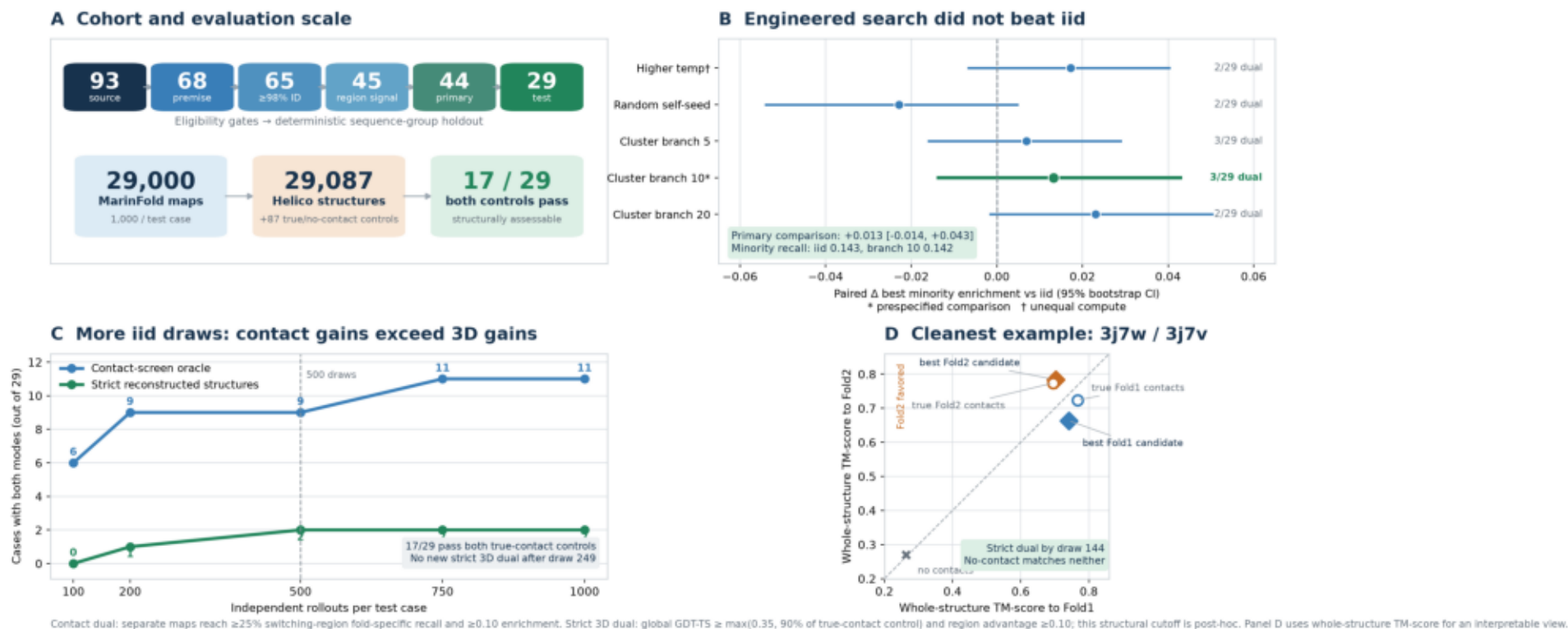


fold_switching_summary.png

One-figure summary of the exp304 fold-switching analysis: cohort and evaluation scale, paired inference-search comparisons, iid contact and structure coverage through 1,000 draws, and the cleanest dual-fold example in whole-structure TM-score space.

Fold-switching inference: rare dual-fold recovery, no demonstrated search gain

29 sequence-group held-out cases • 1,000 iid maps per case • every map folded individually with Helico



Conclusion

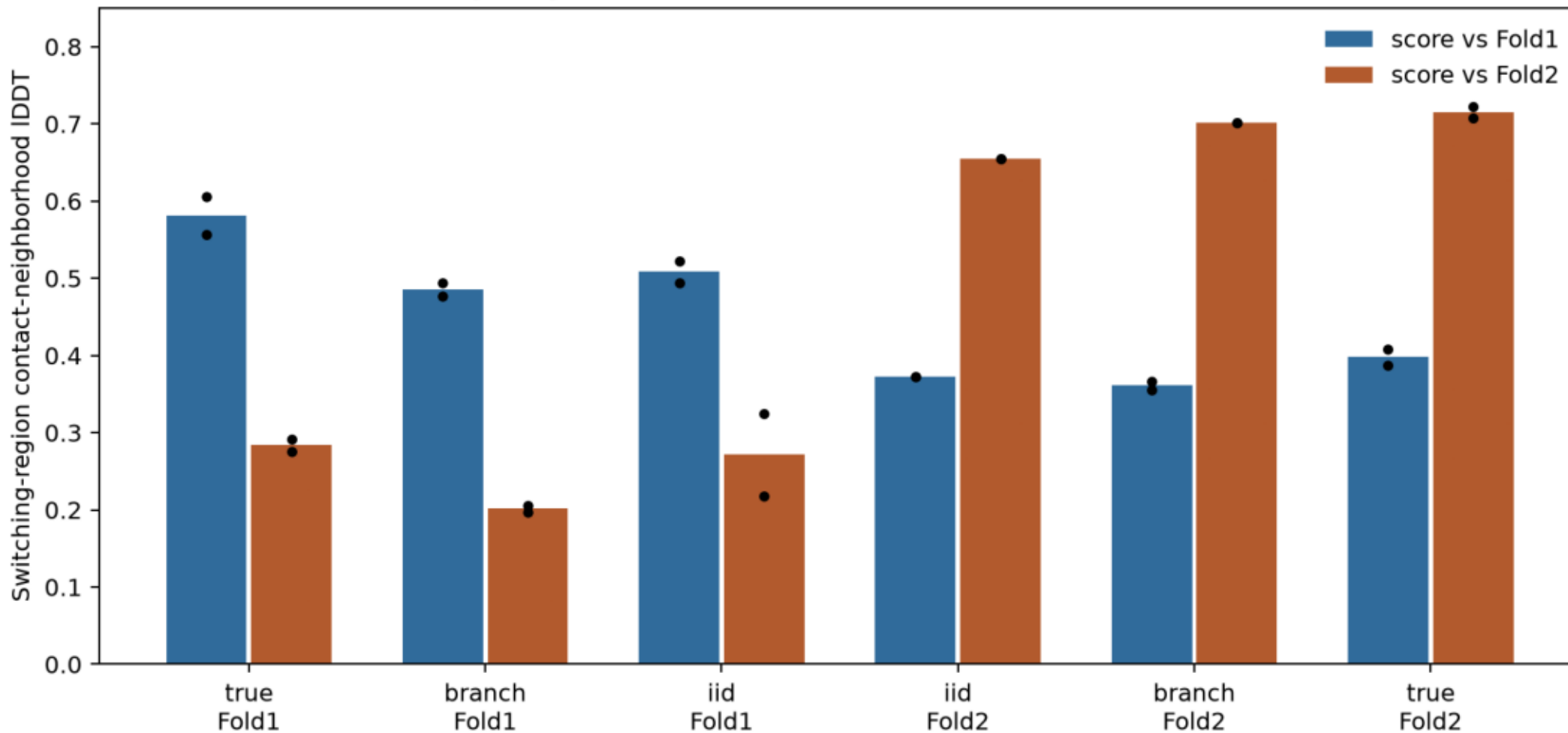
IID rollouts can recover both experimental folds, but rarely: 11/29 contact-screen duals → 2/29 strict structural duals → 1 clean no-contact-controlled case. The tested branching search has no demonstrated gain; draws 500-1,000 added no strict 3D duals.

\$ plot_fold_switching_summary.py

helico_3j7_structural_check.png

Helico structures from two reference input chains; black points are the two inputs. Both 200-rollout shortlist methods yield fold-specific switching-region structures on this protein. True-contact controls show the attainable separation for this reconstruction model.

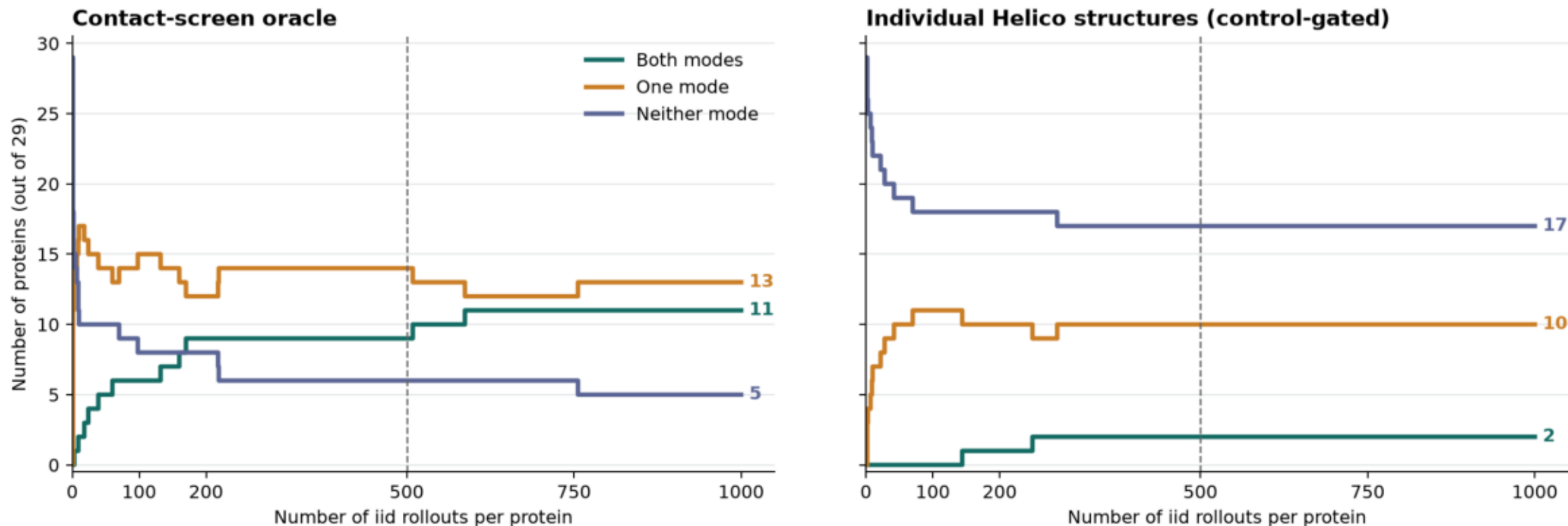
3j7w/3j7v: both blind selectors recover two local folds



helico_iid_structural_coverage.png

Cumulative oracle coverage of two fold modes under the contact-map screen and under cross-reference structural scoring of every individual Helico output. The structural thresholds are shown on the figure; raw cross-reference Kabsch GDT-TS and CA RMSD values are retained for threshold sensitivity analyses.

Does iid sampling reveal both experimentally observed folds?

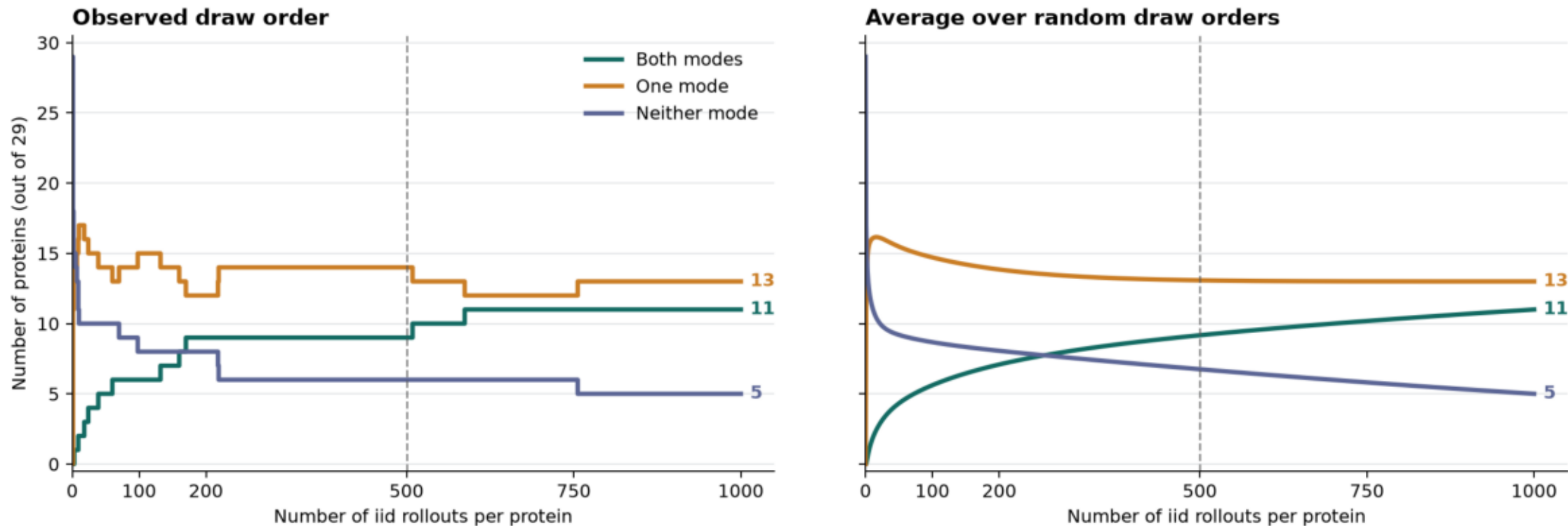


Post-hoc structural hit: common-position Kabsch GDT-TS $\geq \max(0.35, 90\% \text{ of the true-contact control})$ and switching-region advantage ≥ 0.10 .
Each Helico prediction uses one individual MarFold map; both controls pass for 17/29 proteins; dashed line: 500 draws.

iid1000_mode_coverage.png

Cumulative oracle contact-mode coverage among 29 primary test proteins. Left: exact sampled order; right: mean over permutations of the observed 1,000 maps for each protein. Fold-like means a reference-aware contact screening hit, not a verified alternate 3D structure.

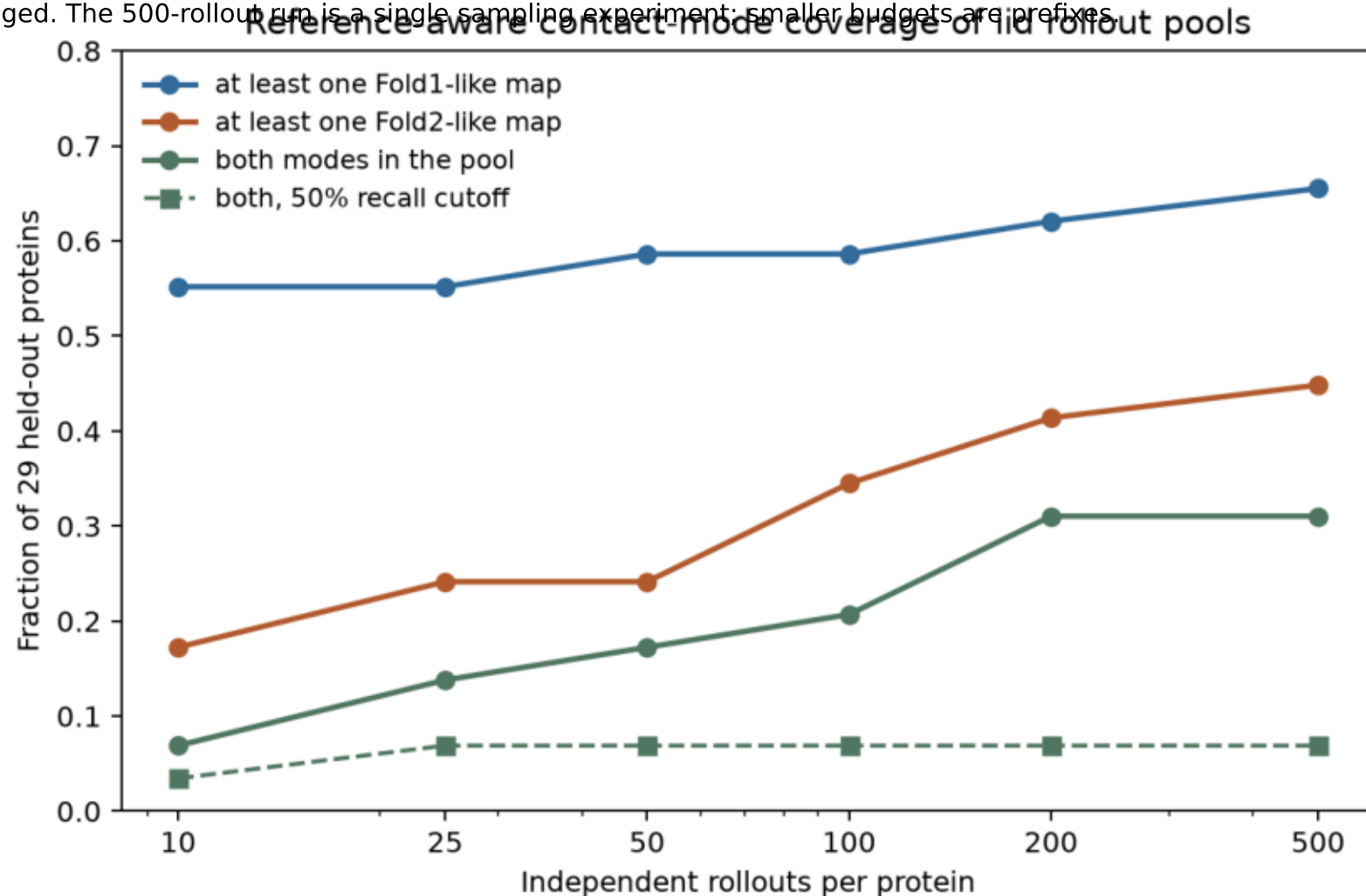
Does more iid sampling reveal both fold-like contact modes?



Each protein is in exactly one category. These are reference-aware contact hits, not verified 3D folds. Dashed line: previous 500-draw limit.
Right panel averages permutations of these same 1,000 maps; it is not a forecast beyond 1,000.

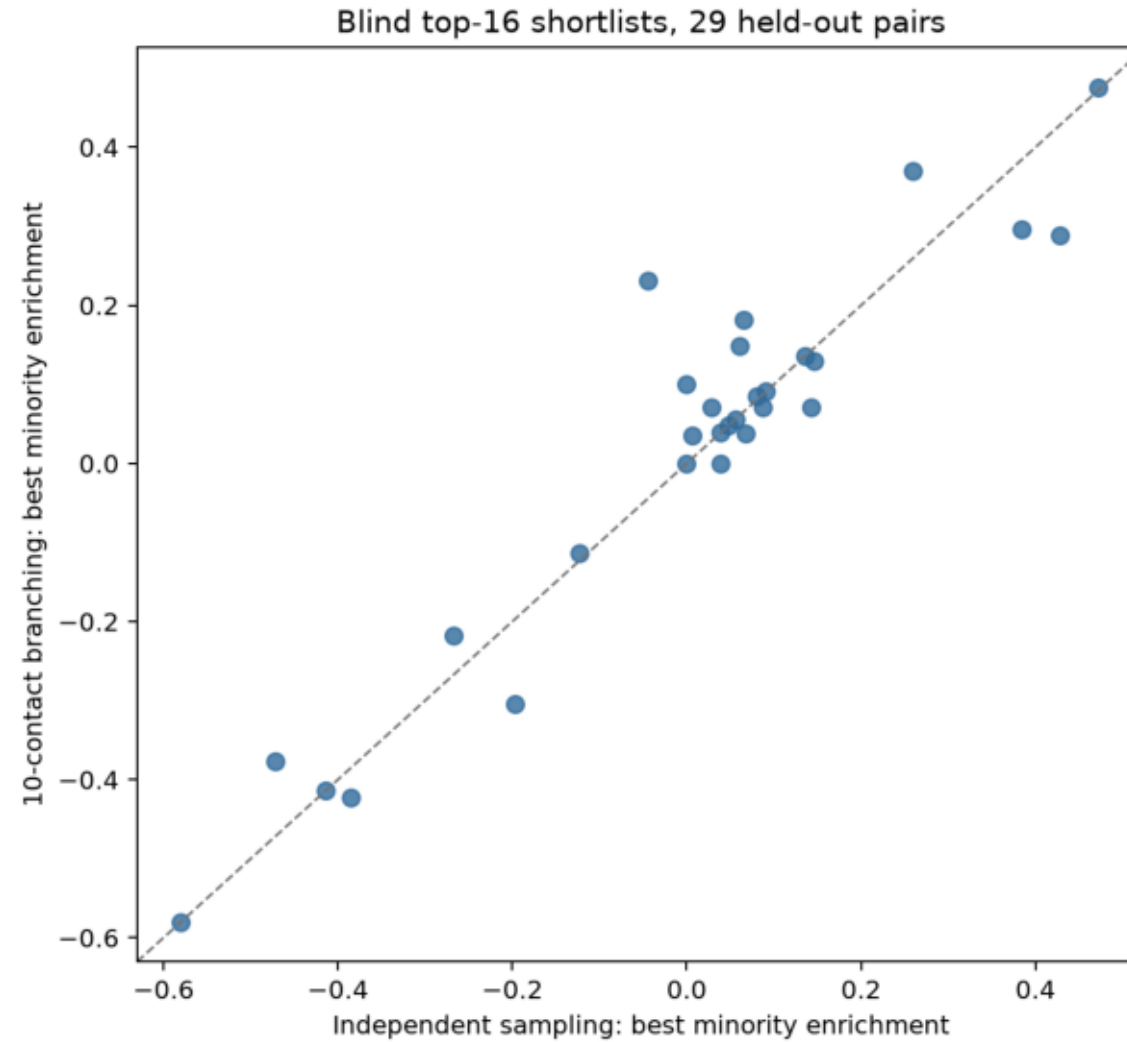
iid_mode_coverage.png

All curves inspect every map with reference contacts; none is a blind selection rate. Main hit cutoff: at least 10 switching-region predictions, 25% unique-contact recall, and 0.10 enrichment versus the opposing fold. Dashed curve uses 50% recall with the other conditions unchanged. The 500-rollout run is a single sampling experiment; smaller budgets are prefixes.



paired_enrichment.png

Each point is one held-out primary fold-switching pair. Both methods use the same 100 root rollouts plus 100 arm rollouts; the diagonal is parity.

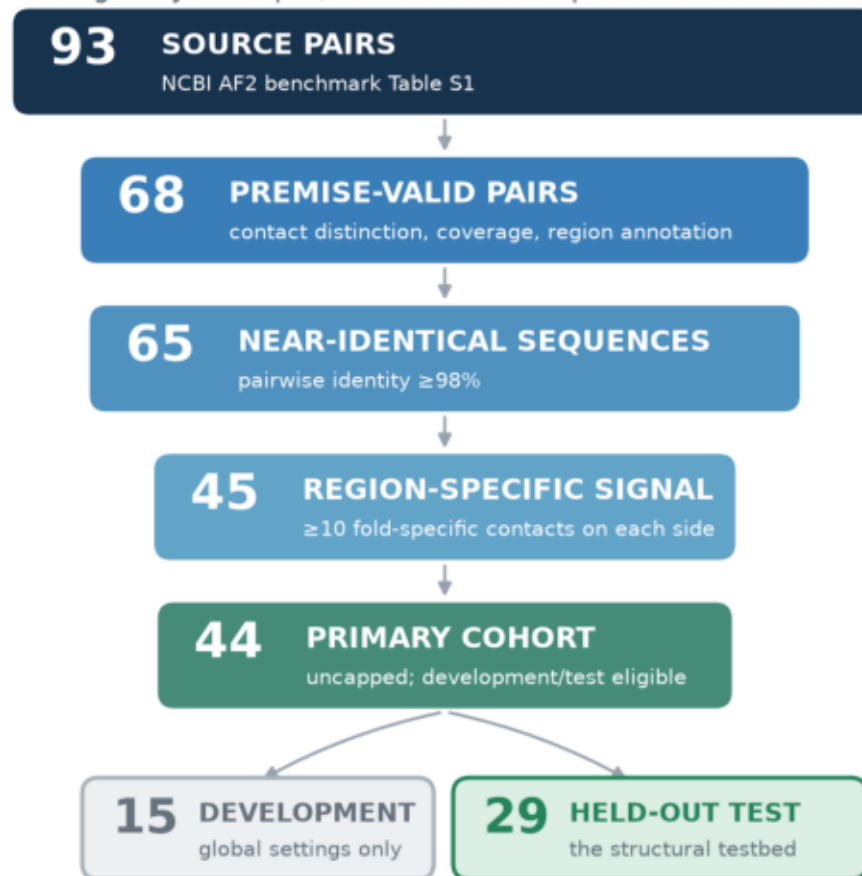


selection_funnel.png

Selection of the 29-pair held-out fold-switching testbed from the 93 literature-curated AF2 benchmark reference pairs. Counts are reconstructed from the exp301 premise gate and the frozen exp304 cohort.

How the 29-protein fold-switching test set was selected

Cohort eligibility and split, frozen before exp304 held-out accuracy was inspected



EXCLUSIONS AND SPLIT

–25 premise failures

17: <10 global unique contacts on one side
14: <50% common resolved-chain coverage
3: switch region not located
Reason counts overlap.

–3 sequence variants

Below 98% identity; retained in secondary analyses.

–20 weak regional signal

Fewer than 10 fold-specific contacts touching the annotated switching region for at least one fold.

–1 context-capped

4zt0c_4cmqb, L=1,338; only 6% of rollouts finished under the 8,192-token context budget.

Sequence-group holdout

43 groups at ≥30% identity and ≥50% either-direction coverage. SHA-256 group order: 15 development, 29 test; no group crosses.

Final 29: 17 exact-sequence pairs; 12 have 1-5 substitutions.

No fold-match, Helico, GDT, RMSD, or TM-score outcome entered selection.